Voice Biometrics

Data acquisition, Feature extraction - Characteristics

What Is Voice Biometrics and Why Does It Matter?

Voice biometrics is the automated process of recognizing or verifying a person's identity by analyzing unique vocal characteristics derived from the physiological structure of the vocal tract and learned behavioral patterns of speech.

Speaker Identification	Speaker Verification	Speaker Diarization
Determines WHO is speaking from a set of known speakers (closed-set) or unknown speakers (open-set). No prior claim needed.	Confirms whether a speaker IS who they claim to be. A 1:1 match between claimed identity and voice model.	Segments audio to determine WHEN each speaker is talking — "who spoke when" in multi-speaker recordings.

Technical Characteristics of Voice

Physiological Characteristics

- **Vocal Tract Shape**
  - Length and geometry of pharynx, larynx, oral/nasal cavities determine resonant frequencies (formants).
- **Glottal Source**
  - Vibration pattern of vocal folds determines fundamental frequency (F0) and voice quality (breathiness, roughness).
- **Articulators**
  - Tongue, lips, teeth, jaw shape co-articulation patterns creating speaker-specific pronunciation habits.
- **Subglottal System**
  - Lung capacity and bronchial resonances subtly affect voice energy distribution across frequencies.

Behavioral Characteristics

- **Prosody & Rhythm**
  - Speaking rate, syllable duration, stress patterns, and intonation contours are highly speaker-specific.
- **Lexical Preferences**
  - Word choice, vocabulary richness, and filler words (" uh", "like") form a linguistic fingerprint.
- **Accent & Dialect**
  - Regional and social accents produce consistent phoneme substitution and intonation patterns.
- **Dynamic Range**
  - Habitual loudness variation, pause frequency, and breath patterns contribute to voice identity.

Acoustic Properties of the Voice

Fundamental Frequency (F0)

The rate at which the vocal folds vibrate. Males: ~120 Hz avg; Females: ~210 Hz avg. Carries intonation (pitch) information.

80–260 Hz

Formant Frequencies (F1–F4)

Resonance peaks of the vocal tract. F1 & F2 differentiate vowels; F3 & F4 carry personal voice quality. Highly speaker-discriminative.

200–4000 Hz

Bandwidth & Duration

Formant bandwidth reflects vocal tract damping. Speaking rate and phoneme duration patterns are behavioral biometric indicators.

Spectral spread

Spectral Envelope

Smoothed spectral shape captures vocal tract transfer function. The basis for LPC and cepstral analysis in speaker recognition.

Overall shape

Voice Source Quality

Harmonic-to-noise ratio (HNR), jitter, shimmer quantify glottal vibration regularity — key indicators of voice health and identity.

H1–H2, HNR

Microintonation

Subtle, semi-random F0 fluctuations unique to each speaker. Captured by micro-prosodic analysis in high-accuracy systems.

Fine F0 variation

Data Acquisition: Capture High:
Quality Voice Signals for Biometric Analysis

Speech Source → Microphone / Transducer → Analog Preprocessing → ADC Sampling → Digital Buffer → Feature Pipeline

Microphone Types

Dynamic mics: robust in high SPL; Condenser mics: wider frequency response, ideal for controlled environments; MEMS mics: used in mobile devices and IoT; Directional (cardioid, supercardioid) vs omnidirectional patterns affect noise rejection.

Sampling & Quantization

Standard: 8 kHz / 8-bit (telephony PESQ bandwidth 300–3400 Hz). High fidelity: 16–48 kHz / 16-bit. Nyquist theorem requires $F_s \geq 2 \times F_{max}$. Anti-aliasing low-pass filter must precede ADC. Dithering reduces quantization artifacts.

Recording Environments

Controlled lab (anechoic chamber) vs real-world noisy conditions. SNR >20 dB preferred. Room acoustics introduce reverberation (RT60). Multi-channel arrays enable beamforming for noise suppression and spatial filtering.

Channel Variability

Different microphones, codecs (AMR, G.711, Opus), and transmission channels alter spectral characteristics. Channel compensation (CMVN, RASTA filtering, domain adaptation) is essential for cross-channel robustness.

Signal Pre-processing: Preparing Raw Audio for Feature Extraction

1. **Pre-emphasis**
   Apply high-pass filter $H(z) = 1 - \alpha z^{-1}$ ($\alpha = 0.97$) to boost high-frequency spectral components attenuated by the glottal source model, improving spectral flatness for analysis.

2. **Framing**
   Segment continuous speech into short overlapping frames (20–30 ms frames, 10 ms hop). Speech is quasi-stationary within frames — short enough to assume stationarity, long enough to estimate spectrum reliably.

3. **Windowing**
   Apply analysis window (Hamming, Hanning, or Blackman) to each frame to reduce spectral leakage caused by discontinuities at frame boundaries. Hamming window: $w(n) = 0.54 - 0.46 \cdot \cos(2\pi n/N)$.

4. **Voice Activity Detection (VAD)**
   Distinguish speech frames from silence/noise. Methods: energy thresholding, zero-crossing rate, LSTM-based VAD. Essential to prevent noise frames from contaminating speaker models.

5. **Noise Reduction**
   Spectral subtraction, Wiener filtering, MMSE-STSA estimators remove stationary additive noise. Deep learning denoisers (RNNoise, DeepFilterNet) handle non-stationary noise in real-world conditions.

6. **Cepstral Mean Variance Normalization (CMVN)**
   Normalize cepstral feature vectors to zero mean and unit variance over the utterance. Removes convolutional channel noise (microphone/transmission effects) — critical for cross-channel robustness.

Feature Extraction: Transforming Audio signals into Discriminative Speaker Representation

- *Feature extraction converts preprocessed speech frames into compact numerical vectors that maximally discriminate between speakers while remaining robust to non-speaker variability (noise, channel, speaking style changes).*

Spectral Features	Prosodic Features	Formant Features	Deep Learned Features
• Mel Frequency Cepstral Coefficients (MFCC) • Linear Predictive Coding (LPC) • LP-Derived Cepstrum (LPCC) • Perceptual Linear Prediction (PLP) • Short-Time Fourier Transform (STFT)	• Fundamental Frequency (F0 / Pitch) • Speaking Rate & Rhythm • Energy Contour & Intensity • Intonation Patterns • Jitter & Shimmer	• Formant Frequencies F1–F4 • Formant Bandwidths • Formant Trajectories • Vowel Space Coverage • Formant Dispersion	• d-vectors (DNN embeddings) • i-vectors (total variability) • x-vectors (TDNN embeddings) • Speaker Embeddings (Wav2Vec) • Contrastive / Self-supervised

MFCC — Mel Frequency Cepstral Coefficients: The Gold Standard Feature for Voice Biometrics

Frame (20ms) → Hamming Window → FFT (512-pt) → Mel Filter Bank → Log Energy → DCT → MFCC Vector

Mel Scale

Perceptually motivated frequency scale: $m = 2595 \cdot \log_{10}(1 + f/700)$. Mimics the human auditory system's non-linear frequency resolution — higher resolution at low frequencies, lower at high frequencies.

Filter Bank Design

Typically 20–40 triangular filters spaced on Mel scale from 0 Hz to Nyquist. Each filter computes weighted sum of spectral energy in its band, producing a Mel spectrum vector representing timbral quality.

Discrete Cosine Transform

Applied to log Mel spectrum to decorrelate filter bank outputs and produce cepstral coefficients. DCT concentrates information in first 12–13 coefficients (C0–C12), which are most discriminative.

Delta & Delta-Delta

First-order (Δ) and second-order (ΔΔ) temporal derivatives appended to static MFCC vector. Capture dynamic spectral change over time. Standard feature vector: 13 MFCC + 13Δ + 13ΔΔ = 39 dimensions.

Liftering

Cepstral liftering (weighting higher cepstral coefficients) improves performance by compensating for the natural decay of cepstral coefficients with order. Sinusoidal lifter with L=22 is standard.

Augmentations

Energy coefficient (C0), pitch-augmented MFCC, shifted delta cepstrum (SDC), and RASTA-PLP are common variants. SDC captures medium-term spectral dynamics useful for language/dialect ID.

LPC, LPCC & Other Spectral Features: Model-Based and Perceptual Feature Extraction Methods

Linear Predictive Coding (LPC)

- Models each speech sample as a linear combination of p previous samples: $s(n) = \Sigma a_k \cdot s(n-k) + Gu(n)$
- Prediction coefficients $\{a_k\}$ encode the spectral envelope of the vocal tract transfer function.
- Typical order p = 10–16 for telephone-bandwidth speech.
- All-pole filter model $H(z) = G / A(z)$ captures formant resonances efficiently.
- Highly efficient computation via autocorrelation method (Levinson-Durbin recursion).
- LPC spectrum is smooth, ideal for formant estimation and vocal tract modeling.

LP-Derived Cepstrum (LPCC)

- Cepstral coefficients derived from LPC: $c_n = a_n + (1/n)\Sigma(k=1..n-1)\ k \cdot c_k \cdot a_{n-k}$
- More robust than raw LPC coefficients; decorrelated representation of vocal tract.
- Better speaker discrimination than LPC due to reduced inter-coefficient correlation.
- Less sensitive to spectral tilt compared to MFCC in some conditions.
- Commonly used in early GMM-based speaker recognition systems.
- Liftering applied to LPCC improves discrimination across different phoneme contexts.

Perceptual Linear Prediction (PLP)

- Extends LPC with perceptual processing: equal-loudness weighting, Bark-scale filter bank, intensity-loudness compression (cubic root).
- Bark scale approximates basilar membrane frequency resolution more accurately than Mel scale.
- All-pole model fitted to perceptually weighted Bark spectrum.
- More robust to recording level variation than MFCC due to loudness normalization.
- Useful in noisy and telephone conditions; competitive with MFCC in many tasks.
- RASTA-PLP adds temporal filtering to remove slow channel variations.

Feature Comparison: MFCC — most popular, perceptual, efficient | LPCC — vocal tract model-based, less correlated | PLP — level-robust, Bark-based, RASTA-filtered

Formant Features & Fundamental Frequency: Vocal tract Resonances and Pitch as Biometric identifiers

FORMANT FREQUENCIES

F1 (300–900 Hz)

Inversely related to tongue height. Rises for open vowels (/a/), falls for close vowels (/i/, /u/). Reflects oral cavity opening.

F2 (700–2500 Hz)

Related to tongue backness. High F2 for front vowels (/i/), low for back vowels (/u/). Most vowel-discriminative formant.

F3 (2000–3500 Hz)

Related to lip rounding and tongue tip. Important for rhotic /r/ and lateral /l/. Highly speaker-specific across all phonemes.

F4 (3000–4500 Hz)

Reflects pharyngeal cavity properties. Less phoneme-dependent than F1–F3; primary carrier of individual vocal tract character.

Formant Dispersion

Mean spacing between formants correlates with vocal tract length — longer tracts (males) have lower mean formant frequencies.

FUNDAMENTAL FREQUENCY (F0)

F0 Definition

F0 is the rate of vocal fold vibration (GHz). Perceptually perceived as pitch. Adult males: 80–180 Hz; Females: 150–300 Hz; Children: 200–450 Hz.

Pitch Extraction

Algorithms: AMDF, RAPT, PYIN, CREPE (deep learning). Accuracy depends on voicing detection and noise level. Unvoiced speech has no F0.

Jitter (Freq. Perturbation)

Cycle-to-cycle F0 variation (%) unique to each speaker's vocal fold mechanics. RAP jitter, PPQ, and DDP jitter are clinical and biometric measures.

Shimmer (Amp. Perturbation)

Cycle-to-cycle amplitude variation of glottal pulses. APQ and dB shimmer reflect vocal fold surface irregularity — speaker-specific property.

Prosodic Pitch Patterns

Long-term F0 statistics (mean, SD, range, shape of intonation contour) are behavioral biometrics that encode speaker's language and emotional style.

Speaker Modeling & Matching, ()

GMM-UBM

1995–2010

Gaussian Mixture Model trained on Universal Background Model. Speaker model = MAP-adapted GMM from UBM. Score = log-likelihood ratio. Dominant method for decades.

i-Vector / PLDA

2010–2016

Total variability space maps utterance GMM statistics to low-dim i-vector (400-dim). PLDA backend models speaker and channel subspaces for scoring. Significant accuracy leap.

x-Vector / TDNN

2017–2020

Time Delay Neural Network extracts x-vectors. Statistical pooling over time aggregates frame-level features. Trained with classification loss. Outperforms i-vectors significantly.

End-to-End / Self-supervised

2020–Present

Wav2Vec 2.0, WavLM, HuBERT encoders pre-trained on massive unlabeled data. Fine-tuned for speaker ID. Near-human EER on VoxCeleb benchmarks.

Equal Error Rate (EER) Progress on VoxCeleb1: GMM-UBM: ~10% | i-Vector: ~5% | x-Vector: ~2.5% | End-to-End: ~0.8%

Applications & Security Considerations: Real world Deployment and Antispoofing Measures

Banking & Finance

Phone banking authentication, fraud detection, transaction verification. ING, HSBC, Nuance deploy voice biometrics for call center authentication.

Law Enforcement

Speaker recognition in forensic casework, surveillance audio analysis, court admissible voice comparisons (LR-based framework).

Smart Devices & IoT

Voice assistant personalization (Alexa, Siri), smart home access control, multi-user speaker separation and recognition.

Healthcare

Patient identification, medication compliance monitoring, detecting cognitive/neurological disorders from voice characteristics.

⚠ Anti-Spoofing & Liveness Detection — Critical Security Layer

Replay Attack

Recording played back. Counter: Liveness detection, challenge-response, acoustic environment mismatch detection.

Voice Conversion

Shift speaker's voice to target. Counter: Artifact detection in spectral statistics, classifier on voice conversion artifacts.

TTS Synthesis

High-quality synthesized voice. Counter: Anti-spoofing model (ASVspoof challenge systems) trained on synthesized speech artifacts.

Summary

Voice Uniqueness

Each voice is shaped by unique vocal tract anatomy (physiological) and learned speech habits (behavioral) — making it a robust biometric.

Data Quality is Foundational

Acquisition, preprocessing, and VAD determine the ceiling of system performance. Clean, well-calibrated data is non-negotiable.

MFCC Remains Central

Despite advances in deep learning, MFCC and cepstral features remain foundational; modern systems build upon them.

Feature Fusion Wins

Combining spectral, prosodic, formant, and deep learned features yields the highest accuracy in real-world conditions.

Security is Multi-Layered

Anti-spoofing, liveness detection, and multi-factor authentication must complement voice biometrics in production systems.

Active Research Field

Self-supervised learning, privacy-preserving biometrics, and cross-lingual robustness are active frontiers driving future progress.